

Nice HMS ABDM APIs — Error Reporting Reference

Audience: external integrators consuming the Nice HMS ABDM data-sharing APIs. This document is the authoritative reference for **how errors are returned** so you can refine your error handling and advance your integration. It is not an implementation guide.

1. Base URL

All endpoints are called with an `Authorization: Bearer <idToken>` header (except `/auth_token`, which issues that token). Endpoint slugs are listed in §7.

```
https://asia-south1-psychic-city-328609.cloudfunctions.net/<endpoint>
```

- Default method is **POST**.
- `/discharge_patient` is **PUT**.
- `/health_document` is **multipart/form-data** (POST). Do not set a `Content-Type` header manually — use the multipart mode of your client so it emits the boundary; a hand-set `Content-Type` (or more than one) has no boundary and the request is rejected.

2. Error envelope (every non-2xx response)

Every error response uses the same JSON shape. There are no other error shapes.

```
{
  "message": "Human-readable summary of the failure",
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request body",
    "details": [
      { "field": "patient.abhaAddress", "message": "The direct input of Abha Address or Abha number is not possible." }
    ]
  }
}
```

Rules

Field	Always present?	Meaning
message	Yes	Top-level summary (kept for backward compatibility — safe to keep parsing it).
error.code	Yes	Stable, machine-readable code. Match on this. See §4.
error.message	Yes	Curated, human-readable message.
error.details	No — only on VALIDATION_ERROR	Array of { field, message }, one entry per failed field.

Example — non-validation error

```
{
  "message": "Patient not found",
  "error": {
    "code": "PATIENT_NOT_FOUND",
    "message": "Patient not found"
  }
}
```

Example — validation error

```
{
  "message": "Invalid request body",
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Invalid request body",
    "details": [
      { "field": "patient.mobile", "message": "Invalid mobile number" },
      { "field": "patient.dob", "message": "Date of birth must be a valid past date" }
    ]
  }
}
```

3. Response headers

Every response — success or error — includes the same headers.

x-request-id

- If your request included an `x-request-id` header, its value is **echoed back** unchanged.
- Otherwise a new UUID is generated.
- The value lives in the **header only** — never inside the JSON body.

Send an `x-request-id` on every request and log the request/response pair against it. When escalating an issue to Nice HMS support, include this value — it correlates to the server-side log of the full (unredacted) error.

Access-Control-Allow-Origin

Every response includes `Access-Control-Allow-Origin: *`, so the API can be called directly from browser-based clients (CORS is enabled for all origins).

4. Error code taxonomy

Match on `error.code` — these are stable strings.

<code>error.code</code>	HTTP	Meaning
<code>VALIDATION_ERROR</code>	400	Request body failed schema validation. See <code>error.details</code> for field-level reasons. Also used for an unsupported upload type, a missing file, or a PDF combined with images on <code>health_document</code> .

VALIDATION_ERROR	413	An uploaded file exceeds the 5 MB limit (health_document).
INVALID_EMAIL_FORMAT	400	An email string is malformed.
INVALID_DOCTOR_GCP_FHIR_ID	400	A doctorGcpId does not exist in the FHIR store (Practitioner).
DOCTOR_DETAILS_REQUIRED	400	doctorDetails is missing (health_document).
INVALID_REQUEST_METHOD	400	Wrong HTTP method for the endpoint.
UNAUTHORIZED	401	Missing/invalid/expired token, or the token is not mapped to an organization.
SUBSCRIPTION_EXPIRED	403	The organization's subscription is invalid or expired.
PATIENT_NOT_FOUND	404	No patient matches the supplied patientId / abhaAddress for this organization.
ENCOUNTER_NOT_FOUND	404	The supplied encounterId was not found for this patient/organization.
DOCTOR_NOT_FOUND	404	The Practitioner FHIR id does not exist (edit_doctor).
NO_ACTIVE_ENCOUNTER	409	The patient has no encounter at all (not even a finished one), so a document cannot be attached to a care context.
EMAIL_ALREADY_EXISTS	409	Duplicate email when creating a doctor.
ABDM_GATEWAY_ERROR	502	Upstream ABDM gateway returned an error (e.g. OTP verification failed, auth mode unavailable).
LINK_TOKEN_MISSING	502	ABDM care-context link token was not found after the callback — a genuine failure, not a rate limit.
LINK_TOKEN_RATE_LIMITED	429	ABDM refused to issue a link token because its cap of 3 new link tokens per ABHA address per 24 hours was reached. Reuse the existing token (valid ~6 months) and retry after 24 hours.
DB_ERROR	500	A database operation failed. Curated generic message.
INTERNAL_ERROR	500	Any unhandled / unknown error. Curated generic message.

5. HTTP status code summary

Status	Category
200	Success
400	Bad request / validation
401	Unauthorized
403	Forbidden (subscription)
404	Not found
409	Conflict (duplicate email) / no active encounter

413	Payload too large (uploaded file exceeds the 5 MB limit)
429	Rate limited (ABDM link-token daily cap reached)
500	Internal server error
502	Bad gateway (ABDM upstream)

6. Validation semantics (important)

1. **Extra / unknown fields are ignored (stripped), not rejected.** You will not get an error for sending fields the API doesn't know about.
2. **Validation runs before auth.** A request that is both unauthenticated *and* has an invalid body will return `400 VALIDATION_ERROR` first (auth is checked after body validation). Plan your client accordingly.
3. **Field paths** in `details[].field` use dot notation, e.g. `patient.firstName` , `doctorDetails.0.doctorGcpId` , `patient.dob` .
4. **Curated messages only.** Responses never leak stack traces, SQL, DB error text, GCP FHIR paths/IDs, or raw ABDM payloads. The full unredacted cause is logged server-side against the `x-request-id` .

7. Per-route error reference

"Shared codes" (present on every **authenticated** route) are:

`VALIDATION_ERROR (400)` · `UNAUTHORIZED (401)` · `SUBSCRIPTION_EXPIRED (403)` · `DB_ERROR (500)` · `INTERNAL_ERROR (500)`

Each route below lists the **shared codes + its own additional codes**.

Where a route says "**patientNeeds**", the body must include at least one of `abhaAddress` (string) or `patientId` (number). Omitting both yields `400 VALIDATION_ERROR` with the message *"Either abhaAddress or patientId is required"*.

7.1 /auth_token — POST (public, no token)

Issues the Firebase ID token used to authenticate every other endpoint.

Request fields

- `email` — valid email (required)
- `password` — non-empty (required)

Error codes

- `VALIDATION_ERROR (400)` — malformed body
- `UNAUTHORIZED (401)` — invalid email or password
- `DB_ERROR (500)` · `INTERNAL_ERROR (500)`

Notes: Not authenticated; does not check subscription.

7.2 /create_patient — POST (auth)

Creates a **non-ABHA** patient (MySQL + GCP FHIR).

Request fields — patient object

- `firstName` — required, non-empty
- `gender` — "Male" | "Female" | "Other"
- `dob` — YYYY-MM-DD , a valid **past** date
- `mobile` — exactly 10 digits
- `state` — required, non-empty
- `district` — required, non-empty
- `abhaAddress` / `abhaNumber` — **rejected** (see note)
- optional passthrough: `lastName` , `middleName` , `prefix` , `address` , `email` , `pincode` , `pan` , `adhar` , `phone` , `internalid` , etc.

Error codes

- Shared codes only (no route-specific extras).

Notes: ABHA patients must be created via `/patient_register_abha_otp_init` + `/patient_register_abha_otp_confirm`. Passing `abhaAddress` or `abhaNumber` here returns `400 VALIDATION_ERROR` with message *"The direct input of Abha Address or Abha number is not possible."*

7.3 /create_doctor — POST (auth)

Creates a doctor (Practitioner in FHIR + MySQL user).

Request fields

- `user.email` — valid email
- `user.firstName` , `user.lastName` — required
- `user.mobile` — 10 digits
- `doctorPrintName` , `medicalLicenseNumber` , `specialty` , `qualification` — required
- `gender` — "Male" | "Female" | "Other"
- `abdmProfessionalId` — optional

Error codes

- `VALIDATION_ERROR` (400)
 - `INVALID_EMAIL_FORMAT` (400) — backstop for malformed email
 - `EMAIL_ALREADY_EXISTS` (409) — email already registered
 - shared codes
-

7.4 /edit_doctor — POST (auth)

Updates an existing doctor's Practitioner resource.

Request fields

- `doctorGcpFhirId` — required
- `doctorPrintName` , `medicalLicenseNumber` , `specialty` , `qualification` — required
- `gender` — "Male" | "Female" | "Other"
- `abdmProfessionalId` — optional

Error codes

- `VALIDATION_ERROR` (400)
- `DOCTOR_NOT_FOUND` (404) — `doctorGcpFhirId` does not exist in the FHIR store

- shared codes
-

7.5 /get_doctor_opd_patients_by_date — POST (auth)

Lists a doctor's OPD patients for a date.

Request fields

- date — required
- doctorGcpId — required

Error codes: shared codes only.

Note: orgId in the body (if present) is ignored — the organization is taken from the token.

7.6 /get_doctor_ipd_patients_by_date — POST (auth)

Lists a doctor's IPD (in-patient) patients.

Request fields

- doctorGcpId — required
- date — optional (no-op)

Error codes: shared codes only.

Note: orgId in the body (if present) is ignored.

7.7 /patient_by_id — POST (auth)

Fetches a single patient by MySQL id.

Request fields

- patientId — number (required)

Error codes

- PATIENT_NOT_FOUND (404)
 - shared codes
-

7.8 /patient_by_abhaAddress — POST (auth)

Fetches a single patient by ABHA address.

Request fields

- abhaAddress — required

Error codes

- PATIENT_NOT_FOUND (404)
 - shared codes
-

7.9 /patient_register_abha_otp_init — POST (auth)

Step 1 of ABHA registration: search ABHA address / request OTP.

Request fields

- `abhaAddress` — required
- `authMode` — "MOBILE_OTP" | "AADHAAR_OTP" | "DONT_KNOW" . `MOBILE_OTP` sends the OTP to the mobile registered against the ABHA in ABDM; `AADHAAR_OTP` sends it to the Aadhaar-linked mobile; `DONT_KNOW` sends nothing and returns the modes the ABHA supports. Repeat the same mode in `/patient_register_abha_otp_confirm` .

Error codes

- `VALIDATION_ERROR` (400)
 - `ABDM_GATEWAY_ERROR` (502) — auth mode not available or blocked for this ABHA, or ABDM rejected the request. `error.message` carries ABDM's own reason.
 - shared codes
-

7.10 `/patient_register_abha_otp_confirm` — POST (auth)

Step 2 of ABHA registration: verify OTP and create the patient.

Request fields

- `transactionId` — required
- `otp` — required
- `abhaAddress` — required
- `authMode` — optional, "MOBILE_OTP" | "AADHAAR_OTP" , defaults to "MOBILE_OTP" . Must match the mode used in `/patient_register_abha_otp_init` .

Error codes

- `VALIDATION_ERROR` (400)
 - `ABDM_GATEWAY_ERROR` (502) — OTP verification failed / no user returned; or the OTP was requested on a different `authMode` . `error.message` carries ABDM's own reason.
 - shared codes
-

7.11 `/opd_patient` — POST (auth)

Creates an OPD encounter for a patient.

Request fields

- `date` — required (ISO date)
- `doctorDetails` — array of { `doctorName`, `doctorGcpId` } , min 1
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400) — a `doctorGcpId` is not a valid Practitioner
 - `PATIENT_NOT_FOUND` (404)
 - shared codes
-

7.12 `/admit_patient` — POST (auth)

Creates an IPD admission encounter.

Request fields

- `doa` — required (ISO date)
- `doctorDetails` — array of `{ doctorName, doctorGcpId }`, min 1
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400)
- `PATIENT_NOT_FOUND` (404)
- shared codes

7.13 `/discharge_patient` — PUT (auth)

Discharges a patient (closes the encounter).

Request fields

- `encounterId` — required
- `dod` — required (ISO date)
- `doctorDetails` — optional
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400)
- `PATIENT_NOT_FOUND` (404)
- `ENCOUNTER_NOT_FOUND` (404) — encounter id not found
- shared codes

Document endpoints — encounter resolution. The five document endpoints below (`/discharge_summary` , `/diagnostic_report` , `/op_consultation` , `/procedure_ot_notes` , `/health_document`) all accept an optional `encounterId` . When supplied it must exist for the given patient **and** belong to the caller's organization. When omitted, the document is attached to the patient's open encounter, or — if there is none — to their most recent encounter (even a finished one). If the patient has no encounter at all, the request fails with `409 NO_ACTIVE_ENCOUNTER` .

7.14 `/discharge_summary` — POST (auth)

Creates a DischargeSummary composition and pushes it to ABDM.

Request fields

- `doctorDetails` — array of `{ doctorName, doctorGcpId }`, min 1
- `text` , `status` , `date` — required
- `compositionId` , `encounterId` — optional
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400)
- `PATIENT_NOT_FOUND` (404) · `ENCOUNTER_NOT_FOUND` (404) · `NO_ACTIVE_ENCOUNTER` (409)
- `ABDM_GATEWAY_ERROR` (502) · `LINK_TOKEN_MISSING` (502) · `LINK_TOKEN_RATE_LIMITED` (429)
- shared codes

7.15 /diagnostic_report — POST (auth)

Creates a DiagnosticReport composition and pushes it to ABDM.

Request fields

- `performer` — array of { `doctorName`, `doctorGcpId` }, min 1
- `testName` — required, non-empty
- `category` — required; one of "Hematology" | "Biochemistry" | "Microbiology" | "Radiology" | "Others"
- `text`, `status` — required, non-empty
- `requester`, `conclusion`, `compositionId`, `date`, `encounterId` — optional
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400) — validates `performer` **and** `requester`
- `PATIENT_NOT_FOUND` (404) · `ENCOUNTER_NOT_FOUND` (404) · `NO_ACTIVE_ENCOUNTER` (409)
- `ABDM_GATEWAY_ERROR` (502) · `LINK_TOKEN_MISSING` (502) · `LINK_TOKEN_RATE_LIMITED` (429)
- shared codes

Notes: `date` is optional and defaults to the request timestamp when omitted.

7.16 /op_consultation — POST (auth)

Creates an OP Consultation composition and pushes it to ABDM.

Request fields

- `doctorDetails` — array of { `doctorName`, `doctorGcpId` }, min 1
- `date`, `status` — required
- optional: `compositionId`, `encounterId`, `chiefComplaints`, `medicalHistory`, `physicalExamination`, `medicines[]`, `opdProcedure`, `followUp`
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400)
 - `PATIENT_NOT_FOUND` (404) · `ENCOUNTER_NOT_FOUND` (404) · `NO_ACTIVE_ENCOUNTER` (409)
 - `ABDM_GATEWAY_ERROR` (502) · `LINK_TOKEN_MISSING` (502) · `LINK_TOKEN_RATE_LIMITED` (429)
 - shared codes
-

7.17 /procedure_ot_notes — POST (auth)

Creates a procedure/OT-notes composition and pushes it to ABDM.

Request fields

- `doctorDetails` — array of { `doctorName`, `doctorGcpId` }, min 1
- `date`, `status` — required
- optional: `compositionId`, `encounterId`, `opdProcedure`, `followUp`
- `patientNeeds`

Error codes

- `INVALID_DOCTOR_GCP_FHIR_ID` (400)

- PATIENT_NOT_FOUND (404) · ENCOUNTER_NOT_FOUND (404) · NO_ACTIVE_ENCOUNTER (409)
- ABDM_GATEWAY_ERROR (502) · LINK_TOKEN_MISSING (502) · LINK_TOKEN_RATE_LIMITED (429)
- shared codes

7.18 /health_document — POST (multipart/form-data, auth)

Uploads a scanned health document (PDF, JPEG or PNG) and creates a HealthDocumentRecord composition.

Request fields (form fields)

- doctorDetails — **JSON string** (required), e.g. `"[{\"doctorName\": \"Dr Umesh\", \"doctorGcpId\": \"...\"}]\"`
- text — required; document title, e.g. Prescription , Lab report
- status — required; one of preliminary , final , amended , entered-in-error
- date , compositionId , encounterId — optional
- abhaAddress or patientId — at least one (patientNeeds)

Files

- page1 — required (PDF, JPEG or PNG, ≤ 5 MB)
- page2 , page3 , page4 — optional (PDF, JPEG or PNG, ≤ 5 MB)
- **A PDF must be uploaded on its own**, in any one of the four fields; a PDF cannot be combined with images.
- A file is typed by its own bytes, not by the part's declared Content-Type . Any other format (DOCX, TIFF, HEIC, ...) is rejected.

ABDM notification

- Only status: "final" is pushed to ABDM. Any other value stores the composition without notifying ABDM, and the 200 response reports abdmNotified: false with abdmNotifiedReason .
- A file larger than 5 MB is rejected with 413 — it is never truncated.

Error codes

- INVALID_REQUEST_METHOD (400) — non-POST
- DOCTOR_DETAILS_REQUIRED (400) — doctorDetails missing
- VALIDATION_ERROR (400) — invalid form fields; unsupported file type; no file supplied; a PDF combined with images
- VALIDATION_ERROR (413) — a file exceeds the 5 MB limit
- INVALID_DOCTOR_GCP_FHIR_ID (400)
- PATIENT_NOT_FOUND (404) · ENCOUNTER_NOT_FOUND (404) · NO_ACTIVE_ENCOUNTER (409)
- ABDM_GATEWAY_ERROR (502) · LINK_TOKEN_MISSING (502) · LINK_TOKEN_RATE_LIMITED (429)
- shared codes

8. Not covered here

- /uhi and /uhi/whatsapp follow the **UHI protocol** (search/init/confirm/cancel/status), which has its own acknowledgement and error format — documented separately, not via this envelope.

9. Quick reference — which code for which mistake

You did this	You'll see
--------------	------------

Missing/invalid token	401 UNAUTHORIZED
Expired subscription	403 SUBSCRIPTION_EXPIRED
Bad field / wrong enum / missing required field	400 VALIDATION_ERROR (+ details)
A file over 5 MB on health_document	413 VALIDATION_ERROR
An unsupported file type, no file, or a PDF combined with images on health_document	400 VALIDATION_ERROR
A file on health_document you expected to reach ABDM, but status isn't final	200, with abdmNotified: false
Wrong doctorGcpId	400 INVALID_DOCTOR_GCP_FHIR_ID
Unknown patientId / abhaAddress	404 PATIENT_NOT_FOUND
Unknown encounterId, or one that belongs to another patient/org	404 ENCOUNTER_NOT_FOUND
Patient has no encounter at all on a document endpoint	409 NO_ACTIVE_ENCOUNTER
Duplicate email on create_doctor	409 EMAIL_ALREADY_EXISTS
ABDM OTP/search/verify failure	502 ABDM_GATEWAY_ERROR
ABDM link token missing after callback (genuine failure)	502 LINK_TOKEN_MISSING
A 4th+ link-token generate for the same ABHA within 24 hours	429 LINK_TOKEN_RATE_LIMITED
Anything else unexpected	500 INTERNAL_ERROR / 500 DB_ERROR